Sentiment Analysis of Restaurant Reviews Case Study

Growl.

Your stomach makes a loud noise as the feeling of hunger permeates throughout your body. You open your fridge hoping to find something to eat, only to be greeted by empty shelves. With no other option, you pull out your phone and search for nearby restaurants. Instantly, dozens of choices appear. But with so many options — each with hundreds of reviews — how do you know which restaurant is actually worth your time?

This is where sentiment analysis can help.

The Problem: Online reviews play a major role in the restaurant industry. Customers rely on reviews to decide where to eat, while restaurant owners use customer feedback to improve their businesses. Platforms such as Yelp contain thousands of user-generated reviews that provide valuable insight into customer experiences.

However, manually reading and interpreting large volumes of reviews is both time-consuming and inefficient. As the number of reviews grows, identifying meaningful patterns in customer sentiment becomes increasingly difficult without automation.

The Project: This project explores how natural language processing (NLP) and machine learning can be used to classify restaurant reviews based on their associated star ratings. Using the Yelp Restaurant Reviews dataset containing over 20,000 reviews, the project analyzes review text, sentiment, and review length to identify patterns that influence customer satisfaction.

The project aims to demonstrate how machine learning models can transform unstructured review text into meaningful insights. By building a sentiment classification pipeline, this analysis shows how NLP techniques can be applied to real-world data to predict ratings, uncover customer preferences, and better understand the factors that drive positive dining experiences.

Project Goal: The primary goal of this project is to train a machine learning model that maps the textual features of a review to its associated star rating. Through this process, the project aims to evaluate how sentiment and writing patterns influence customer perception and overall restaurant ratings.

Everything you need to get started is in the GitHub Repository below. Good luck and have fun!

GitHub Repository: https://github.com/KC-Night/CS3-Sentiment-Analysis-of-Restaurant-Reviews-Case-Study.git